

MTB 603: Numerical Analysis

Master Examination Solutions and Analytical Guide

Date of Exam: 28/02/2023

Time: 1 Hour

Full Marks (FM): 20

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Official Instructions: Answer Question no. 1 and any two from 2–5. The figures in the right hand margin indicate the marks of questions.

Editorial & Methodological Note

To provide the most thorough, publication-grade academic document, **all five questions** (Question 1 through Question 5) from the examination paper have been fully solved below. Each problem features precise definitions, line-by-line mathematical proofs, tabular iterations, and exact fractional arithmetic.

Question 1

[5 Marks]

Question 1(a)

[3 Marks]

Problem Statement

Define relative error in approximation of a number. Find the relative error in rounding off the number 0.76452 upto three decimal places.

Solution

1. Theoretical Definition of Errors: Let x denote the true (exact) value of a numerical quantity, and let x_A (or x^*) denote its approximate value.

- (i) **Absolute Error (E_a):** The absolute error is defined as the absolute magnitude of the difference between the true value and the approximate value:

$$E_a = \Delta x = |x - x_A|.$$

- (ii) **Relative Error (E_r):** The relative error is defined as the ratio of the absolute error to the magnitude of the true value:

$$E_r = \frac{E_a}{|x|} = \frac{|x - x_A|}{|x|}, \quad (x \neq 0).$$

The relative error is a dimensionless measure of precision, reflecting the fractional discrepancy relative to the true magnitude.

(iii) **Percentage Error (E_p):** The percentage error is the relative error expressed as a percentage:

$$E_p = E_r \times 100\% = \frac{|x - x_A|}{|x|} \times 100\%.$$

2. Rounding Off 0.76452 Upto Three Decimal Places:

- **Given True Value:** $x = 0.76452$.
- **Target Precision:** Three decimal places ($k = 3$).
- The first three decimal digits are .764.
- The fourth decimal digit (the first digit to be discarded) is 5, followed by the non-zero digit 2 (i.e., the discarded part is .00052 $>$ 0.00050).
- Under standard round-to-nearest rules, since the discarded fraction exceeds half a unit in the last retained place, the third decimal digit 4 must be increased by 1:

$$x_A = 0.765.$$

3. Error Calculations:

(a) **Absolute Error (E_a):**

$$E_a = |x - x_A| = |0.76452 - 0.76500| = |-0.00048| = 0.00048 = 4.8 \times 10^{-4}.$$

(b) **Relative Error (E_r):**

$$\begin{aligned} E_r &= \frac{E_a}{x} = \frac{0.00048}{0.76452} \\ &= \frac{48}{76452} = \frac{4}{6371} \\ &\approx 0.000627845 \dots \approx 6.278 \times 10^{-4}. \end{aligned}$$

(c) **Percentage Error (E_p):**

$$E_p = E_r \times 100\% \approx 0.06278\% \approx 0.063\%.$$

Final Answer for Question 1(a)

- **Rounded Number (x_A):** 0.765
- **Absolute Error (E_a):** 0.00048 (4.8×10^{-4})
- **Relative Error (E_r):** ≈ 0.0006278 (6.278×10^{-4})
- **Percentage Error (E_p):** $\approx 0.0628\%$

Question 1(b)

[2 Marks]

Problem Statement

Write down a fixed point iteration scheme to find a real root of the equation $x^3 + x^2 - 1 = 0$ lying in $(0, 1)$.

Solution

1. Root Isolation in $(0, 1)$: Let $f(x) = x^3 + x^2 - 1$. Evaluating $f(x)$ at the interval boundaries:

$$f(0) = 0^3 + 0^2 - 1 = -1 < 0, \quad f(1) = 1^3 + 1^2 - 1 = 1 > 0.$$

Since $f(x)$ is continuous on $[0, 1]$ and $f(0)f(1) < 0$, by the **Intermediate Value Theorem**, there exists a real root $\xi \in (0, 1)$. Moreover, $f'(x) = 3x^2 + 2x > 0$ for all $x \in (0, 1)$, confirming that f is strictly monotonic and the root ξ is unique.

2. Formulation of the Fixed-Point Scheme: To solve $f(x) = 0$ by fixed-point iteration, we rewrite the equation in the form:

$$x = \phi(x),$$

which induces the iteration sequence $x_{n+1} = \phi(x_n)$. For the sequence to converge to the root ξ , the iteration function $\phi(x)$ must satisfy the **contraction condition**:

$$|\phi'(x)| < 1, \quad \forall x \in [0, 1].$$

Rearranging $x^3 + x^2 - 1 = 0$:

$$\begin{aligned} x^2(x+1) - 1 = 0 &\implies x^2(x+1) = 1 \\ &\implies x^2 = \frac{1}{x+1} \\ &\implies x = \frac{1}{\sqrt{x+1}} = (x+1)^{-1/2}. \end{aligned}$$

Thus, we define the iteration function:

$$\phi(x) = \frac{1}{\sqrt{x+1}} = (x+1)^{-1/2}.$$

3. Rigorous Convergence Verification: Differentiating $\phi(x)$ with respect to x :

$$\phi'(x) = -\frac{1}{2}(x+1)^{-3/2} = -\frac{1}{2(x+1)^{3/2}}.$$

For any $x \in [0, 1]$, the magnitude of the derivative is:

$$|\phi'(x)| = \frac{1}{2(x+1)^{3/2}}.$$

Evaluating at the endpoints of $[0, 1]$:

$$\text{At } x = 0: \quad |\phi'(0)| = \frac{1}{2(1)^{3/2}} = \frac{1}{2} = 0.5 < 1,$$

$$\text{At } x = 1: \quad |\phi'(1)| = \frac{1}{2(2)^{3/2}} = \frac{1}{4\sqrt{2}} \approx 0.1768 < 1.$$

Since $(x+1)^{3/2}$ is strictly increasing for $x \geq 0$, the maximum value of $|\phi'(x)|$ occurs at $x = 0$:

$$\max_{x \in [0, 1]} |\phi'(x)| = \frac{1}{2} = 0.5 < 1.$$

Because $|\phi'(x)| \leq 0.5 < 1$ for all $x \in [0, 1]$, the Banach Contraction Mapping Theorem guarantees that the sequence converges to the unique root $\xi \approx 0.75488$ for any starting point $x_0 \in [0, 1]$.

Final Answer for Question 1(b)

The required fixed point iteration scheme is:

$$\mathbf{x}_{n+1} = \frac{1}{\sqrt{\mathbf{x}_n + 1}}, \quad n = 0, 1, 2, \dots$$

with any initial approximation $x_0 \in (0, 1)$ (such as $x_0 = 0.5$ or $x_0 = 0.75$).

Question 2**[5 Marks]****Problem Statement**

By Newton-Raphson find the smallest positive root of the equation $x^3 - 6x + 4 = 0$ correct upto three decimal places.

Solution

1. Analysis and Location of the Roots: Let $f(x) = x^3 - 6x + 4 = 0$. We evaluate $f(x)$ at selected real points:

$$f(0) = 0^3 - 6(0) + 4 = 4 > 0,$$

$$f(1) = 1^3 - 6(1) + 4 = -1 < 0,$$

$$f(2) = 2^3 - 6(2) + 4 = 8 - 12 + 4 = 0.$$

- Since $f(0) = 4 > 0$ and $f(1) = -1 < 0$, a real root lies in the open interval $(0, 1)$.
- Since $f(2) = 0$, $x = 2$ is an exact positive integer root.
- Factoring out $(x - 2)$ from $f(x)$ via polynomial division:

$$x^3 - 6x + 4 = (x - 2)(x^2 + 2x - 2) = 0.$$

The quadratic factor $x^2 + 2x - 2 = 0$ yields:

$$x = \frac{-2 \pm \sqrt{2^2 - 4(1)(-2)}}{2} = \frac{-2 \pm \sqrt{12}}{2} = -1 \pm \sqrt{3}.$$

The three roots of the cubic equation are:

$$x_1 = \sqrt{3} - 1 \approx 0.73205, \quad x_2 = 2.00000, \quad x_3 = -\sqrt{3} - 1 \approx -2.73205.$$

The two positive roots are $\sqrt{3} - 1 \approx 0.732$ and 2.000 . Hence, the **smallest positive root** is:

$$\xi = \sqrt{3} - 1 \approx 0.732051 \dots \in (0, 1).$$

2. Newton-Raphson Iteration Formula: The Newton-Raphson iteration formula is:

$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}.$$

For $f(x) = x^3 - 6x + 4$:

$$f'(x) = 3x^2 - 6.$$

Substituting $f(x)$ and $f'(x)$:

$$x_{n+1} = x_n - \frac{x_n^3 - 6x_n + 4}{3x_n^2 - 6} = \frac{x_n(3x_n^2 - 6) - (x_n^3 - 6x_n + 4)}{3x_n^2 - 6} = \frac{2x_n^3 - 4}{3x_n^2 - 6}.$$

3. Numerical Iterations: Since $f(0) = 4$ and $f(1) = -1$, and $|f(1)| = 1 < 4 = |f(0)|$, the root is much closer to 1 than to 0. We choose the convenient initial guess $x_0 = 0.7$ (or $x_0 = 1.0$). Carrying out the calculations with 6 decimal places:

Iteration 1: For $n = 0$ ($x_0 = 0.7$):

$$\begin{aligned} f(x_0) &= (0.7)^3 - 6(0.7) + 4 = 0.343 - 4.2 + 4 = 0.143, \\ f'(x_0) &= 3(0.7)^2 - 6 = 3(0.49) - 6 = 1.47 - 6 = -4.53, \\ x_1 &= 0.7 - \frac{0.143}{-4.53} = 0.7 + 0.031567 = \mathbf{0.731567}. \end{aligned}$$

Iteration 2: For $n = 1$ ($x_1 = 0.731567$):

$$\begin{aligned} f(x_1) &= (0.731567)^3 - 6(0.731567) + 4 = 0.391515 - 4.389402 + 4 = 0.002113, \\ f'(x_1) &= 3(0.731567)^2 - 6 = 3(0.535190) - 6 = 1.605571 - 6 = -4.394429, \\ x_2 &= 0.731567 - \frac{0.002113}{-4.394429} = 0.731567 + 0.000481 = \mathbf{0.732048}. \end{aligned}$$

Iteration 3: For $n = 2$ ($x_2 = 0.732048$):

$$\begin{aligned} f(x_2) &= (0.732048)^3 - 6(0.732048) + 4 = 0.392289 - 4.392288 + 4 = 0.000013, \\ f'(x_2) &= 3(0.732048)^2 - 6 = 3(0.535894) - 6 = 1.607683 - 6 = -4.392317, \\ x_3 &= 0.732048 - \frac{0.000013}{-4.392317} = 0.732048 + 0.000003 = \mathbf{0.732051}. \end{aligned}$$

Table of Iterations

n	x_n	$f(x_n)$	$f'(x_n)$	x_{n+1}	$ x_{n+1} - x_n $
0	0.700000	+0.143000	-4.530000	0.731567	0.031567
1	0.731567	+0.002113	-4.394429	0.732048	0.000481
2	0.732048	+0.000013	-4.392317	0.732051	0.000003
3	0.732051	+0.000000	-4.392305	0.732051	0.000000

Since $x_2 \approx 0.732$ and $x_3 \approx 0.732$, the iteration has converged to three decimal places.

Final Answer for Question 2

The smallest positive root of $x^3 - 6x + 4 = 0$ correct upto three decimal places is:

$$\mathbf{x \approx 0.732} \quad (\text{Exact value: } \sqrt{3} - 1 \approx 0.732051).$$

Question 3

[5 Marks]

Problem Statement

Prove that the iteration scheme,

$$x_{n+1} = \frac{x_n}{8} \left[6 + \frac{3a}{x_n^2} - \frac{x_n^2}{a} \right]$$

has third order of convergence to the root $\sqrt{a}$.

Solution

1. Formulation of the Iteration Function: Let the iteration scheme be written as $x_{n+1} = \phi(x_n)$, where:

$$\phi(x) = \frac{x}{8} \left[6 + \frac{3a}{x^2} - \frac{x^2}{a} \right] = \frac{3}{4}x + \frac{3a}{8x} - \frac{x^3}{8a}.$$

The target root is $\alpha = \sqrt{a}$, which satisfies $\alpha^2 = a$.

2. Derivative Criterion for Order of Convergence: By the standard theorem on the order of convergence of one-point iteration methods:

An iteration scheme $x_{n+1} = \phi(x_n)$ has order of convergence $p = 3$ to a root α if and only if:

$$\phi(\alpha) = \alpha, \quad \phi'(\alpha) = 0, \quad \phi''(\alpha) = 0, \quad \text{and} \quad \phi'''(\alpha) \neq 0.$$

In that case, the asymptotic error equation is given by:

$$e_{n+1} = \frac{\phi'''(\alpha)}{3!} e_n^3 + O(e_n^4).$$

3. Successive Derivative Calculations and Evaluations at $\alpha = \sqrt{a}$:

Step 1: Verification of Fixed Point:

$$\begin{aligned} \phi(\sqrt{a}) &= \frac{3}{4}\sqrt{a} + \frac{3a}{8\sqrt{a}} - \frac{(\sqrt{a})^3}{8a} \\ &= \frac{3}{4}\sqrt{a} + \frac{3}{8}\sqrt{a} - \frac{a\sqrt{a}}{8a} \\ &= \left(\frac{3}{4} + \frac{3}{8} - \frac{1}{8} \right) \sqrt{a} = \left(\frac{6+3-1}{8} \right) \sqrt{a} = \sqrt{a}. \end{aligned}$$

Thus $\phi(\sqrt{a}) = \sqrt{a}$, confirming $\sqrt{a}$ is a fixed point.

Step 2: First Derivative $\phi'(x)$:

$$\phi'(x) = \frac{d}{dx} \left[\frac{3}{4}x + \frac{3a}{8}x^{-1} - \frac{1}{8a}x^3 \right] = \frac{3}{4} - \frac{3a}{8x^2} - \frac{3x^2}{8a}.$$

Evaluating at $x = \sqrt{a}$ (where $x^2 = a$):

$$\phi'(\sqrt{a}) = \frac{3}{4} - \frac{3a}{8a} - \frac{3a}{8a} = \frac{3}{4} - \frac{3}{8} - \frac{3}{8} = \frac{3}{4} - \frac{6}{8} = 0.$$

Step 3: Second Derivative $\phi''(x)$:

$$\phi''(x) = \frac{d}{dx} \left[\frac{3}{4} - \frac{3a}{8}x^{-2} - \frac{3}{8a}x^2 \right] = \frac{6a}{8x^3} - \frac{6x}{8a} = \frac{3a}{4x^3} - \frac{3x}{4a}.$$

Evaluating at $x = \sqrt{a}$ (where $x^3 = a\sqrt{a}$):

$$\phi''(\sqrt{a}) = \frac{3a}{4(a\sqrt{a})} - \frac{3\sqrt{a}}{4a} = \frac{3}{4\sqrt{a}} - \frac{3}{4\sqrt{a}} = 0.$$

Step 4: Third Derivative $\phi'''(x)$:

$$\phi'''(x) = \frac{d}{dx} \left[\frac{3a}{4}x^{-3} - \frac{3}{4a}x \right] = -\frac{9a}{4x^4} - \frac{3}{4a}.$$

Evaluating at $x = \sqrt{a}$ (where $x^4 = a^2$):

$$\phi'''(\sqrt{a}) = -\frac{9a}{4a^2} - \frac{3}{4a} = -\frac{9}{4a} - \frac{3}{4a} = -\frac{12}{4a} = -\frac{3}{a} \neq 0.$$

4. Asymptotic Error Equation: Let $e_n = x_n - \sqrt{a}$. Expanding $\phi(x_n)$ by Taylor's theorem about $\alpha = \sqrt{a}$:

$$\begin{aligned} x_{n+1} &= \phi(x_n) = \phi(\sqrt{a}) + \phi'(\sqrt{a})e_n + \frac{\phi''(\sqrt{a})}{2!}e_n^2 + \frac{\phi'''(\sqrt{a})}{3!}e_n^3 + O(e_n^4) \\ \implies x_{n+1} - \sqrt{a} &= 0 + 0 + 0 + \frac{-3/a}{6}e_n^3 + O(e_n^4) \\ \implies e_{n+1} &= -\frac{1}{2a}e_n^3 + O(e_n^4). \end{aligned}$$

Taking absolute values and limits as $n \rightarrow \infty$:

$$\lim_{n \rightarrow \infty} \frac{|e_{n+1}|}{|e_n|^3} = \frac{1}{2a} \neq 0.$$

This proves that the order of convergence of the iteration scheme to the root $\sqrt{a}$ is **three** (third order).

5. Alternative Proof (Exact Algebraic Factorization): We can also derive the exact error relation algebraically for all x_n :

$$\begin{aligned} x_{n+1} - \sqrt{a} &= \frac{x_n}{8} \left[6 + \frac{3a}{x_n^2} - \frac{x_n^2}{a} \right] - \sqrt{a} \\ &= \frac{6ax_n^2 + 3a^2 - x_n^4 - 8a\sqrt{a}x_n}{8ax_n} \\ &= -\frac{x_n^4 - 6ax_n^2 + 8a^{3/2}x_n - 3a^2}{8ax_n}. \end{aligned}$$

Factoring the numerator polynomial in terms of $(x_n - \sqrt{a})$:

$$x_n^4 - 6ax_n^2 + 8a^{3/2}x_n - 3a^2 = (x_n - \sqrt{a})^3(x_n + 3\sqrt{a}).$$

Therefore, we obtain the exact algebraic error identity:

$$e_{n+1} = -\frac{x_n + 3\sqrt{a}}{8ax_n}e_n^3.$$

As $x_n \rightarrow \sqrt{a}$:

$$\lim_{x_n \rightarrow \sqrt{a}} \frac{x_n + 3\sqrt{a}}{8ax_n} = \frac{\sqrt{a} + 3\sqrt{a}}{8a\sqrt{a}} = \frac{4\sqrt{a}}{8a\sqrt{a}} = \frac{1}{2a}.$$

Hence:

$$\lim_{n \rightarrow \infty} \frac{|e_{n+1}|}{|e_n|^3} = \frac{1}{2a},$$

which rigorously establishes that the scheme is cubically convergent (order 3).

Conclusion for Question 3

Since $\phi(\sqrt{a}) = \sqrt{a}$, $\phi'(\sqrt{a}) = 0$, $\phi''(\sqrt{a}) = 0$, and $\phi'''(\sqrt{a}) = -\frac{3}{a} \neq 0$, the iteration scheme has **third order of convergence** with asymptotic error constant $C = \frac{1}{2a}$. ■

Question 4

[5 Marks]

Problem Statement

Use three iterations of the Birge-Vieta method to find a real root of the equation $x^3 - 11x^2 + 32x - 22 = 0$ with initial approximation 0.5 (All calculations need to be performed upto five decimal places).

Solution

1. The Birge-Vieta Method Algorithm: The Birge-Vieta method applies Newton-Raphson iteration to a polynomial:

$$P(x) = a_0x^n + a_1x^{n-1} + \cdots + a_n = 0$$

using Horner's synthetic division scheme to compute $P(p)$ and $P'(p)$ simultaneously. For the cubic polynomial:

$$P(x) = x^3 - 11x^2 + 32x - 22 = 0,$$

the coefficients are:

$$a_0 = 1, \quad a_1 = -11, \quad a_2 = 32, \quad a_3 = -22.$$

Given initial approximation:

$$p_0 = 0.5.$$

In each iteration with current estimate p , we compute:

(a) **Value of the Polynomial $P(p)$:**

$$\begin{aligned} b_0 &= a_0 = 1, \\ b_i &= a_i + p \cdot b_{i-1} \quad (i = 1, 2, 3), \\ P(p) &= b_3. \end{aligned}$$

(b) **Value of the Derivative $P'(p)$:**

$$\begin{aligned} c_0 &= b_0 = 1, \\ c_i &= b_i + p \cdot c_{i-1} \quad (i = 1, 2), \\ P'(p) &= c_2. \end{aligned}$$

(c) **Correction and Next Approximation:**

$$\Delta p = -\frac{P(p)}{P'(p)} = -\frac{b_3}{c_2}, \quad p_{\text{new}} = p + \Delta p = p - \frac{b_3}{c_2}.$$

2. Step-by-Step Iterations (upto 5 decimal places):

Iteration 1 ($p_0 = 0.50000$):

- Compute b_i :

$$\begin{aligned} b_0 &= a_0 = 1.00000, \\ b_1 &= a_1 + p_0 b_0 = -11 + 0.5(1.00000) = -10.50000, \\ b_2 &= a_2 + p_0 b_1 = 32 + 0.5(-10.50000) = 32 - 5.25000 = 26.75000, \\ b_3 &= a_3 + p_0 b_2 = -22 + 0.5(26.75000) = -22 + 13.37500 = -8.62500 = P(p_0). \end{aligned}$$

- Compute c_i :

$$c_0 = b_0 = 1.00000,$$

$$c_1 = b_1 + p_0 c_0 = -10.50000 + 0.5(1.00000) = -10.00000,$$

$$c_2 = b_2 + p_0 c_1 = 26.75000 + 0.5(-10.00000) = 26.75000 - 5.00000 = \mathbf{21.75000} = P'(p_0).$$

Synthetic Division Table 1:

$p_0 = 0.50000$	1.00000	-11.00000	32.00000	-22.00000
		+0.50000	-5.25000	+13.37500
b	1.00000	-10.50000	26.75000	$-\mathbf{8.62500} = P(p_0)$
		+0.50000	-5.00000	
c	1.00000	-10.00000	$\mathbf{21.75000} = P'(p_0)$	

$$\Delta p_0 = -\frac{b_3}{c_2} = -\frac{-8.62500}{21.75000} = \frac{8.62500}{21.75000} \approx +0.39655,$$

$$\mathbf{p_1 = p_0 + \Delta p_0 = 0.50000 + 0.39655 = 0.89655.}$$

Iteration 2 ($p_1 = 0.89655$):

- Compute b_i :

$$b_0 = 1.00000,$$

$$b_1 = -11 + (0.89655)(1.00000) = -10.10345,$$

$$b_2 = 32 + (0.89655)(-10.10345) = 32 - 9.05825 = 22.94175,$$

$$b_3 = -22 + (0.89655)(22.94175) = -22 + 20.56843 = \mathbf{-1.43157} = P(p_1).$$

- Compute c_i :

$$c_0 = 1.00000,$$

$$c_1 = -10.10345 + (0.89655)(1.00000) = -9.20690,$$

$$c_2 = 22.94175 + (0.89655)(-9.20690) = 22.94175 - 8.25445 = \mathbf{14.68730} = P'(p_1).$$

Synthetic Division Table 2:

$p_1 = 0.89655$	1.00000	-11.00000	32.00000	-22.00000
		+0.89655	-9.05825	+20.56843
b	1.00000	-10.10345	22.94175	$-\mathbf{1.43157} = P(p_1)$
		+0.89655	-8.25445	
c	1.00000	-9.20690	$\mathbf{14.68730} = P'(p_1)$	

$$\Delta p_1 = -\frac{b_3}{c_2} = -\frac{-1.43157}{14.68730} \approx +0.09747,$$

$$\mathbf{p_2 = p_1 + \Delta p_1 = 0.89655 + 0.09747 = 0.99402.}$$

Iteration 3 ($p_2 = 0.99402$):

- Compute b_i :

$$b_0 = 1.00000,$$

$$b_1 = -11 + (0.99402)(1.00000) = -10.00598,$$

$$b_2 = 32 + (0.99402)(-10.00598) = 32 - 9.94614 = 22.05386,$$

$$b_3 = -22 + (0.99402)(22.05386) = -22 + 21.92200 = -\mathbf{0.07800} = P(p_2).$$

- Compute c_i :

$$c_0 = 1.00000,$$

$$c_1 = -10.00598 + (0.99402)(1.00000) = -9.01196,$$

$$c_2 = 22.05386 + (0.99402)(-9.01196) = 22.05386 - 8.95807 = \mathbf{13.09579} = P'(p_2).$$

Synthetic Division Table 3:

$p_2 = 0.99402$	1.00000	-11.00000	32.00000	-22.00000
		+0.99402	-9.94614	+21.92200
b	1.00000	-10.00598	22.05386	$-\mathbf{0.07800} = P(p_2)$
		+0.99402	-8.95807	
c	1.00000	-9.01196	$\mathbf{13.09579} = P'(p_2)$	

$$\Delta p_2 = -\frac{b_3}{c_2} = -\frac{-0.07800}{13.09579} \approx +0.00596,$$

$$\mathbf{p_3} = p_2 + \Delta p_2 = 0.99402 + 0.00596 = \mathbf{0.99998}.$$

3. Comprehensive Summary of the Three Iterations:

Iteration (k)	p_k	$P(p_k) = b_3$	$P'(p_k) = c_2$	$\Delta p_k = -b_3/c_2$	p_{k+1}
1	0.50000	-8.62500	21.75000	+0.39655	0.89655
2	0.89655	-1.43157	14.68730	+0.09747	0.99402
3	0.99402	-0.07800	13.09579	+0.00596	0.99998

Verification of Exactness

Evaluating $P(1) = 1^3 - 11(1)^2 + 32(1) - 22 = 1 - 11 + 32 - 22 = 0$. Thus, $x = 1$ is an exact integer root of the cubic equation. The sequence of approximations $\{0.50000, 0.89655, 0.99402, 0.99998\}$ rapidly converges to this exact root.

Final Answer for Question 4

After three iterations of the Birge-Vieta method starting from $p_0 = 0.5$, the approximation to the root correct upto five decimal places is:

$$\mathbf{p_3 \approx 0.99998} \quad (\text{Exact root: } 1.00000).$$

Question 5

[5 Marks]

Problem Statement

Solve the following system of equations:

$$\begin{aligned} 2x_1 + x_2 + x_3 + 2x_4 &= 2, \\ 4x_1 + 2x_3 + x_4 &= 3, \\ 3x_1 + 2x_2 + 2x_3 &= -1, \\ x_1 + 3x_2 + 2x_3 &= -4, \end{aligned}$$

using Gauss elimination method with partial pivoting (Do not use approximations).

Solution

The directive “**Do not use approximations**” requires all calculations to be carried out in exact rational arithmetic (fractions).

1. Initial Augmented Matrix Formulation: Writing the system $A\mathbf{x} = \mathbf{b}$ in augmented matrix format $[A \mid \mathbf{b}]$:

$$\left[\begin{array}{cccc|c} 2 & 1 & 1 & 2 & 2 \\ 4 & 0 & 2 & 1 & 3 \\ 3 & 2 & 2 & 0 & -1 \\ 1 & 3 & 2 & 0 & -4 \end{array} \right] \begin{matrix} (R_1) \\ (R_2) \\ (R_3) \\ (R_4) \end{matrix}$$

2. Step-by-Step Elimination with Partial Pivoting:**Stage 1: Column 1 Pivoting and Elimination**

- **Pivot Selection:** Inspect column 1 entries in all rows:

$$|a_{11}| = 2, \quad |a_{21}| = 4, \quad |a_{31}| = 3, \quad |a_{41}| = 1.$$

The maximum absolute value is in Row 2 ($|a_{21}| = 4$).

- **Row Interchange ($R_1 \longleftrightarrow R_2$):**

$$\left[\begin{array}{cccc|c} 4 & 0 & 2 & 1 & 3 \\ 2 & 1 & 1 & 2 & 2 \\ 3 & 2 & 2 & 0 & -1 \\ 1 & 3 & 2 & 0 & -4 \end{array} \right]$$

- **Row Operations:**

(i) $R_2 \leftarrow R_2 - \left(\frac{2}{4}\right) R_1 = R_2 - \frac{1}{2} R_1$:

$$\begin{aligned} R_2 &\leftarrow [2, 1, 1, 2 \mid 2] - \frac{1}{2}[4, 0, 2, 1 \mid 3] \\ &= \left[0, 1, 1 - 1, 2 - \frac{1}{2} \mid 2 - \frac{3}{2}\right] = \left[0, 1, 0, \frac{3}{2} \mid \frac{1}{2}\right]. \end{aligned}$$

$$(ii) R_3 \leftarrow R_3 - \left(\frac{3}{4}\right) R_1:$$

$$\begin{aligned} R_3 &\leftarrow [3, 2, 2, 0 \mid -1] - \frac{3}{4}[4, 0, 2, 1 \mid 3] \\ &= \left[0, 2, 2 - \frac{3}{2}, 0 - \frac{3}{4} \mid -1 - \frac{9}{4}\right] = \left[0, 2, \frac{1}{2}, -\frac{3}{4} \mid -\frac{13}{4}\right]. \end{aligned}$$

$$(iii) R_4 \leftarrow R_4 - \left(\frac{1}{4}\right) R_1:$$

$$\begin{aligned} R_4 &\leftarrow [1, 3, 2, 0 \mid -4] - \frac{1}{4}[4, 0, 2, 1 \mid 3] \\ &= \left[0, 3, 2 - \frac{1}{2}, 0 - \frac{1}{4} \mid -4 - \frac{3}{4}\right] = \left[0, 3, \frac{3}{2}, -\frac{1}{4} \mid -\frac{19}{4}\right]. \end{aligned}$$

After Stage 1, the matrix is:

$$\left[\begin{array}{cccc|c} 4 & 0 & 2 & 1 & 3 \\ 0 & 1 & 0 & 3/2 & 1/2 \\ 0 & 2 & 1/2 & -3/4 & -13/4 \\ 0 & 3 & 3/2 & -1/4 & -19/4 \end{array} \right]$$

Stage 2: Column 2 Pivoting and Elimination

- **Pivot Selection:** Inspect column 2 entries in rows 2, 3, 4:

$$|a_{22}| = 1, \quad |a_{32}| = 2, \quad |a_{42}| = 3.$$

The maximum absolute value is in Row 4 ($|a_{42}| = 3$).

- **Row Interchange** ($R_2 \longleftrightarrow R_4$):

$$\left[\begin{array}{cccc|c} 4 & 0 & 2 & 1 & 3 \\ 0 & \mathbf{3} & 3/2 & -1/4 & -19/4 \\ 0 & 2 & 1/2 & -3/4 & -13/4 \\ 0 & 1 & 0 & 3/2 & 1/2 \end{array} \right]$$

- **Row Operations:**

$$(i) R_3 \leftarrow R_3 - \left(\frac{2}{3}\right) R_2:$$

$$\begin{aligned} R_3 &\leftarrow \left[0, 2, \frac{1}{2}, -\frac{3}{4} \mid -\frac{13}{4}\right] - \frac{2}{3}\left[0, 3, \frac{3}{2}, -\frac{1}{4} \mid -\frac{19}{4}\right] \\ &= \left[0, 0, \frac{1}{2} - 1, -\frac{3}{4} + \frac{1}{6} \mid -\frac{13}{4} + \frac{19}{6}\right] \\ &= \left[0, 0, -\frac{1}{2}, \frac{-9+2}{12} \mid \frac{-39+38}{12}\right] = \left[0, 0, -\frac{1}{2}, -\frac{7}{12} \mid -\frac{1}{12}\right]. \end{aligned}$$

$$(ii) R_4 \leftarrow R_4 - \left(\frac{1}{3}\right) R_2:$$

$$\begin{aligned} R_4 &\leftarrow \left[0, 1, 0, \frac{3}{2} \mid \frac{1}{2}\right] - \frac{1}{3}\left[0, 3, \frac{3}{2}, -\frac{1}{4} \mid -\frac{19}{4}\right] \\ &= \left[0, 0, 0 - \frac{1}{2}, \frac{3}{2} + \frac{1}{12} \mid \frac{1}{2} + \frac{19}{12}\right] \\ &= \left[0, 0, -\frac{1}{2}, \frac{18+1}{12} \mid \frac{6+19}{12}\right] = \left[0, 0, -\frac{1}{2}, \frac{19}{12} \mid \frac{25}{12}\right]. \end{aligned}$$

After Stage 2, the matrix is:

$$\left[\begin{array}{cccc|c} 4 & 0 & 2 & 1 & 3 \\ 0 & 3 & 3/2 & -1/4 & -19/4 \\ 0 & 0 & -1/2 & -7/12 & -1/12 \\ 0 & 0 & -1/2 & 19/12 & 25/12 \end{array} \right]$$

Stage 3: Column 3 Pivoting and Elimination

- **Pivot Selection:** Inspect column 3 entries in rows 3 and 4:

$$|a_{33}| = \left| -\frac{1}{2} \right| = \frac{1}{2}, \quad |a_{43}| = \left| -\frac{1}{2} \right| = \frac{1}{2}.$$

The pivot candidates are equal in absolute magnitude, so no row swap is needed.

- **Row Operation** ($R_4 \leftarrow R_4 - (1)R_3$):

$$\begin{aligned} R_4 &\leftarrow \left[0, 0, -\frac{1}{2}, \frac{19}{12} \mid \frac{25}{12} \right] - \left[0, 0, -\frac{1}{2}, -\frac{7}{12} \mid -\frac{1}{12} \right] \\ &= \left[0, 0, 0, \frac{19}{12} - \left(-\frac{7}{12} \right) \mid \frac{25}{12} - \left(-\frac{1}{12} \right) \right] \\ &= \left[0, 0, 0, \frac{26}{12} \mid \frac{26}{12} \right] = \left[0, 0, 0, \frac{13}{6} \mid \frac{13}{6} \right]. \end{aligned}$$

The final upper triangular augmented matrix $[U \mid \mathbf{c}]$ is:

$$\left[\begin{array}{cccc|c} 4 & 0 & 2 & 1 & 3 \\ 0 & 3 & 3/2 & -1/4 & -19/4 \\ 0 & 0 & -1/2 & -7/12 & -1/12 \\ 0 & 0 & 0 & 13/6 & 13/6 \end{array} \right]$$

3. Back-Substitution Process:

Step 1: From Row 4 for x_4 :

$$\frac{13}{6}x_4 = \frac{13}{6} \implies \mathbf{x_4 = 1}.$$

Step 2: From Row 3 for x_3 :

$$-\frac{1}{2}x_3 - \frac{7}{12}x_4 = -\frac{1}{12}.$$

Substituting $x_4 = 1$:

$$\begin{aligned} -\frac{1}{2}x_3 - \frac{7}{12} &= -\frac{1}{12} \implies -\frac{1}{2}x_3 = -\frac{1}{12} + \frac{7}{12} = \frac{6}{12} = \frac{1}{2} \\ \implies x_3 &= \frac{1/2}{-1/2} \implies \mathbf{x_3 = -1}. \end{aligned}$$

Step 3: From Row 2 for x_2 :

$$3x_2 + \frac{3}{2}x_3 - \frac{1}{4}x_4 = -\frac{19}{4}.$$

Substituting $x_3 = -1$ and $x_4 = 1$:

$$\begin{aligned} 3x_2 + \frac{3}{2}(-1) - \frac{1}{4}(1) &= -\frac{19}{4} \implies 3x_2 - \frac{6}{4} - \frac{1}{4} = -\frac{19}{4} \\ \implies 3x_2 - \frac{7}{4} &= -\frac{19}{4} \\ \implies 3x_2 &= -\frac{19}{4} + \frac{7}{4} = -\frac{12}{4} = -3 \\ \implies \mathbf{x_2} &= \mathbf{-1}. \end{aligned}$$

Step 4: From Row 1 for x_1 :

$$4x_1 + 0x_2 + 2x_3 + x_4 = 3.$$

Substituting $x_3 = -1$ and $x_4 = 1$:

$$\begin{aligned} 4x_1 + 2(-1) + 1 &= 3 \implies 4x_1 - 2 + 1 = 3 \\ &\implies 4x_1 - 1 = 3 \\ &\implies 4x_1 = 4 \implies \mathbf{x_1 = 1.} \end{aligned}$$

4. Verification in All Original Equations:

$$\text{Equation 1: } 2(1) + 1(-1) + 1(-1) + 2(1) = 2 - 1 - 1 + 2 = 2 \quad (\text{Satisfied } \checkmark)$$

$$\text{Equation 2: } 4(1) + 0(-1) + 2(-1) + 1(1) = 4 - 2 + 1 = 3 \quad (\text{Satisfied } \checkmark)$$

$$\text{Equation 3: } 3(1) + 2(-1) + 2(-1) + 0(1) = 3 - 2 - 2 = -1 \quad (\text{Satisfied } \checkmark)$$

$$\text{Equation 4: } 1(1) + 3(-1) + 2(-1) + 0(1) = 1 - 3 - 2 = -4 \quad (\text{Satisfied } \checkmark)$$

All four equations are satisfied exactly.

Final Answer for Question 5

The exact solution to the linear system obtained via Gauss elimination with partial pivoting is:

$$\mathbf{x_1 = 1, \quad x_2 = -1, \quad x_3 = -1, \quad x_4 = 1.}$$

Master Examination Summary Table

Question	Marks	Final Solution / Result Summary
1(a)	3	Absolute Error = 0.00048, Relative Error $\approx 6.278 \times 10^{-4}$ ($E_p \approx 0.0628\%$).
1(b)	2	Fixed point iteration scheme: $\mathbf{x_{n+1}} = \frac{1}{\sqrt{\mathbf{x_n}+1}}$ ($ \phi'(x) \leq 0.5 < 1$ on $[0, 1]$).
2	5	Smallest positive root of $x^3 - 6x + 4 = 0$ is $\mathbf{x \approx 0.732}$ (exact: $\sqrt{3} - 1$).
3	5	$\lim_{n \rightarrow \infty} \frac{ e_{n+1} }{ e_n ^3} = \frac{1}{2a} \neq 0 \implies$ Third order convergence (cubic convergence).
4	5	Birge-Vieta iterations: $p_1 = 0.89655$, $p_2 = 0.99402$, $\mathbf{p_3 \approx 0.99998}$ (≈ 1.00000).
5	5	Exact rational solution: $\mathbf{x_1 = 1, \quad x_2 = -1, \quad x_3 = -1, \quad x_4 = 1.}$